

PUSHKAR RAJ

+91-7257989026 | pjha2608@gmail.com | LinkedIn | GitHub | LeetCode

Projects

ArthAI

2026

Next.js, FastAPI, PostgreSQL, BullMQ, Redis, Gemini API GitHub | Live Demo

- Architected a multi-agent SaaS platform orchestrating **Gemini API** agents with **Groq/Llama 3.3 70B** fallback for automated audit-report generation; designed scalable **FastAPI** microservices achieving under **500ms** response times.
- Built a three-stage **BullMQ** background-job pipeline on **Redis** and a **RAG** retrieval pipeline with **pgvector**/text-embedding-004 for semantic search; engineered **PostgreSQL** schemas with **Prisma ORM** and compound indexing, and automated **Puppeteer**-based PDF reports with **CI/CD** on Render.

MockMind

2026

Next.js, PostgreSQL (Prisma, Neon), LangGraph, Groq, Vapi, JWT GitHub | Live Demo

- Built a voice-based AI mock-interview platform delivering live spoken interviews via **Vapi** WebRTC (STT/TTS) across Behavioral, Technical, System Design, and HR/Culture Fit formats.
- Designed a **LangGraph** multi-node state-machine – separate nodes for answer evaluation, strategy decisions, and question generation – reducing repetition and hallucination; integrated **Groq LLaMA 3.3 70B** for low-latency inference and **HMAC**-signed tab-close session authentication.

ExpertBook

2026

React.js, Node.js, MongoDB, Socket.io, JWT GitHub | Live Demo

- Built responsive booking dashboards in **React.js** with **JWT**-based authentication and role-based authorization; implemented real-time **WebSocket** updates via **Socket.io** at under **100ms** latency.
- Applied **MongoDB** transactions for concurrency-safe reservations with rollback on failure; automated Zoom/Google Meet link generation and ICS calendar exports on booking confirmation.

PromptWatch

2026

Next.js 15 (Edge Middleware), Upstash Redis, Supabase, Vercel Live Demo

- Engineered an **Edge Middleware** layer intercepting every request pre-page-load, detecting AI crawler traffic (GPTBot, ClaudeBot, PerplexityBot) via User-Agent signature matching, and logging hits to **Upstash Redis**.
- Built a dashboard reading Redis counters through **Supabase**-authenticated REST APIs for live traffic visibility; extending detection into a policy control panel (allow/throttle/block).

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, Java, SQL, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Socket.io, Microservices

Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM, pgvector

AI / ML: Scikit-learn, HuggingFace Transformers, OpenCV, Embeddings, Semantic Search

Frameworks: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Multi-Agent Systems, Prompt Engineering

Developer Tools: Git, GitHub, Postman, VS Code

Cloud & DevOps: Docker, CI/CD, Vercel, Render, Netlify

Messaging & Caching: Redis, BullMQ, Background Jobs

Authentication & APIs: JWT, HMAC, REST APIs, WebSockets

Software Engineering: Data Structures, Algorithms, OOP, System Design, Distributed Systems, Clean Architecture

Open Source / Independent Development

Full Stack Developer and Open Source Contributor

Sep 2023 – Present

Independent Technical Projects

Bengaluru, Karnataka

- Solved **250+** Data Structures and Algorithms problems on LeetCode (arrays, trees, graphs, dynamic programming) with an **80%** solve rate; mentored **50+** students through MERN stack workshops covering REST APIs, frontend integration, and deployment workflows.
- Maintained **15+** GitHub repositories with **150+** commits, enforcing CI/CD pipelines, semantic versioning, structured code reviews, and technical documentation across full-stack and ML projects.

Education

JSS Academy of Technical Education

Sep 2023 – Sep 2027

Bachelor of Engineering, Computer Science Engineering (AI and ML)

Bengaluru, Karnataka

Achievements / Certifications

- **1st Place**, Nexathon 2025 | Dr. T. Thimmaiah Institute of Technology
- Participant, THINK2IMPACT Ideathon 2.0 | AICTE IC-AISMART, JSSATE Bengaluru
- Full Stack Web Development | Udemy, 2024
- Java Programming Masterclass | Apna College, 2025